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Onset of End-Stage Atrophic Age-Related Macular Degeneration as an End Point—A Delphi Study: Classification of Atrophy Meetings Report 7

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Purpose: To investigate whether consensus can be reached on the acceptability of end-stage atrophy onset as a clinical end point in early intervention trials of age-related macular degeneration (AMD), and the criteria for defining such an end point.

Design: A modified Delphi study.

Participants: International panel of experts in AMD, retinal imaging, and histopathology that are part of the Classification of Atrophy Meetings group.

Methods: A modified Delphi study was undertaken to determine if there is consensus on the acceptability of the onset of end-stage atrophic AMD as a clinical end point to evaluate early interventions and the criteria for defining such an end point. Two initial rounds of online surveys were conducted. Aggregate results and anonymized comments were provided after each round, followed by a face-to-face meeting before a final survey round was completed.

Main Outcome Measures: Statements where consensus was reached, defined as ≥80% of responses within the 3-point bracket for agreement or disagreement based on a 9-point Likert rating scale, from a total of 33 statements included in the final round of the survey.

Results: Consensus was reached for the statement that the onset of end-stage atrophic AMD was an appropriate clinical end point to evaluate early interventions (82% responses in agreement). Consensus was also reached for the statement that such an end point should be defined based on anatomical changes that have been previously shown in clinical studies to be associated with marked, but not necessarily complete, functional loss (95% responses in agreement). Consensus was nearly reached for the specific criterion that ≥90% of such atrophic AMD lesions should have ≥1 test location that was ≤10 decibels on 2 microperimetry tests (77% responses in agreement).

Conclusions: There was expert group consensus that the onset of end-stage atrophy is an appropriate clinical end point to evaluate early interventions in AMD, and that such an end point should show evidence of marked functional loss in prior clinical studies. We believe these findings will help to define incident clinical end points that are acceptable to regulatory authorities.

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Geographic atrophy (GA) is the term describing the end-stage atrophic form of nonneovascular age-related macular degeneration (AMD).[1] Geographic atrophy has historically been defined on clinical examination or color fundus photographs as a "*sharply delineated roughly round or oval area of hypopigmentation or depigmentation or apparent absence of the (retinal pigment epithelium [RPE]) in which choroidal vessels are more visible than in surrounding areas that must be at least 175 μm in diameter.*"[1] In more recent years, GA has also been visualized and quantified based on well-demarcated areas of decreased autofluorescence on fundus autofluorescence imaging.[2,3] Geographic atrophy is also seen on OCT as regions with attenuation or loss of the RPE and choroidal

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signal hypertransmission, with evidence of overlying photoreceptor degeneration.[4,5] Given the findings from histopathological analyses that regions of RPE loss secondary to GA show evidence of photoreceptor degeneration,[6–9] it has been assumed that regions with GA are nonseeing, or characterized by an absolute scotoma.[10,11]

The onset of GA has previously been used as a clinical end point in early intervention trials aiming to stop or slow its development.[12–15] However, GA onset has never been used as an end point for regulatory authorities to register an intervention. We are now entering an era where it may be possible to intervene early in the disease process before GA develops. Therefore, we need to assess whether consensus exists within the international AMD clinical community on whether the onset of end-stage atrophic AMD is an acceptable end point, and the criteria for defining such an end point. This consensus would be especially critical to assist regulatory authorities when considering the design of early intervention studies. It will be particularly important to determine if there is consensus on whether structural features alone would be considered sufficient to define such an end point, or if evidence is needed for their association with visual function impairment in either previous clinical studies, or as directly measured within trials. If it is considered necessary that visual function loss should be demonstrated, it would then be important to determine what level of loss would be considered sufficient when defining such an end point.

To address these issues, we designed a modified Delphi study to examine if consensus could be reached on the acceptability and specific criteria for such a clinical end point of end-stage atrophic AMD that could be used in early intervention trials.

Methods

Study Design and Topic Overview

A modified Delphi study was performed among international experts in AMD to examine if consensus could be reached on the acceptability of a clinical end point of atrophic AMD onset to evaluate early interventions, and the criteria for its definition. This end point is intended to encapsulate the end of a disease process with an area of "nonseeing" retina. This study sought to assess the degree of consensus around whether such an end point should be broadly defined based upon anatomical changes alone, anatomical changes associated with visual function loss shown in prior clinical studies, or based on functional loss that is directly measured during the course of an interventional trial. If evidence of functional loss is required, this study also sought to understand broadly what degree of functional loss needs to be demonstrated.

Furthermore, this study sought to understand the application of these broader principles outlined above to a specific scenario in which functional testing was performed using clinical fundus-controlled perimetry ("microperimetry"). This technique measures visual sensitivity at specific locations of the retina and compensates for fixation errors with fundus tracking. This visual function test was chosen as it can be used to capture visual function abnormalities more effectively than commonly used visual function tests such as visual acuity.[16,17] It was also chosen because it has been widely used in clinical trials,[18] and is performed using similar parameters as standard automated perimetry, whose measurements have been considered acceptable by regulatory authorities.[11,19,20]

Survey Methodology

A modified Delphi technique[21,22] was used to examine if consensus could be reached by the Classification of Atrophy Meetings (CAM) group (and not to force consensus) on the study topic, in a similar manner as performed previously by other groups.[23,24] The CAM group comprises an international team of experts in AMD, retinal imaging, and histopathology,[5] whose membership is reviewed prior to each meeting by the chairs of the CAM group (S.R.S., G.S., and F.G.H.) to ensure that those with relevant expertise are invited.

The survey in the first round consisted of 28 statements across 4 sections that were developed by 2 nonvoting cochairs (R.H.G. and Z.W.) and 1 member (S.R.S.) of a subcommittee of the CAM group that led this Delphi study. These statements were derived based on previous discussions by CAM group members and content from previous CAM reports.[5,25–27] They were accompanied by additional analyses of data from previous studies that employed microperimetry testing in individuals with the early stages of AMD and atrophic AMD (described later in this section). An explanation of the background and purpose of this Delphi study and of the study data were provided by the subcommittee cochairs (R.H.G. and Z.W.) through a video recording. A 9-point Likert rating scale was used to indicate the level of agreement or disagreement with each statement by each survey participant, where a score of 1 represents strong disagreement, 5 represents neither agreement nor disagreement, and 9 represents strong agreement. Responses to the statements and the optional provision of free-text comments for each section of the survey were captured electronically and anonymously online.

Results from the first-round survey were used to guide any reformulation of the survey for a second round. Based on these results, 3 additional statements in 1 additional section were added, and the original statements were left unchanged. Participants were provided with the aggregate results and anonymized comments from the first round, as well as methodological clarifications by the subcommittee cochairs, to assist with their responses in the second round, which were again provided anonymously. Aggregate results and anonymized comments from the second round were then presented and discussed in person at a CAM meeting in May 2024. Comments arising from this in-person meeting were thus not anonymous. They were also used to assist the responses in the final round of the survey, which was completed anonymously online by all members who attended this in-person meeting. The results of this final round, completed by 22 members of the CAM group, are presented and summarized in this report.

Supportive Additional Data Analyses Provided

To aid the consideration of the specific level of functional loss that should be seen when defining the onset of end-stage atrophic AMD, a simulation model was developed to provide insights on the microperimetry testing characteristics of a region with a truly nonresponding test location, as reported in detail previously.[28] This simulation was provided to illustrate how the subjective nature of functional testing (where false-positive and false-negative responses occur) can have an impact on how often regions with and without a truly nonresponding test location would meet different functional criteria (such as how often they had a repeatable absolute scotoma). Briefly, this simulation model was developed using data from 128 individuals with bilateral large drusen who underwent microperimetry testing at 6 monthly intervals up to

a 36-month period. Microperimetry was performed using the Macular Integrity Assessment device (CenterVue), using a stimulus pattern consisting of 37 test locations in the central 6° radius. Visual sensitivities were measured with Goldman Size III (0.43°) stimuli that were presented on a mesopic background (1.27 cd/m^2) using a 4-2 staircase strategy. Microperimetry testing results were simulated with and without a truly nonresponding test location. Tests with a truly nonresponding location were simulated by randomly selecting 1 location to have visual sensitivity derived from the distribution of responses from a test location at the optic nerve head (i.e., a location that was expected to be truly nonresponding) from 60 healthy individuals.[28]

These data were then complemented by further analyses of a clinical cohort, as described previously,[29] that underwent targeted, high-density microperimetry testing of regions with at least incomplete RPE and outer retinal atrophy. Testing was performed in a similar manner as described above (using the same device and test parameters), but using a high-density stimulus pattern with 37 test locations sampling a 3.5° diameter region that was manually centered on an atrophic lesion-of-interest.[29] These findings sought to provide insights on how often regions meeting different current definitions of atrophic AMD would meet various functional criteria on clinical microperimetry testing. These findings were not intended to be definitive, as they were derived from a study conducted at a single site and based on grading from a single group. However, these findings served to simply provide information to aid the considerations by the participants of this Delphi study.

A representative example of each of the 3 functional criteria examined for the microperimetry test results from the simulation model and clinical cohort is shown in Figure 1. Findings from repeat testing are summarized in Table 1, as shown to the participants of the Delphi study. Briefly, Table 1 shows, for example, that only 60% of regions with a truly nonresponding test location had ≥1 test location with repeatable absolute scotoma (or <0 decibels [dB]), while approximately 90% of such regions would have ≥1 test location with a repeatable deep visual sensitivity defect (≤10 dB). Note that the cut-off of ≤10 dB was chosen based on previous findings that this value represents the floor of the effective dynamic range of this device. This cut-off for the effective dynamic range is defined as the highest value for which 5% of the retest values are recorded as ≤0 dB, or in other words, statistically indistinguishable from the floor of the physical dynamic range of this device.[30] However, only 17% and 82% of regions with GA as conventionally defined on color fundus photographs that are ≥175 μm in diameter[1] had ≥1 location that was <0 and ≤10 dB on both tests, respectively.

Note that the studies used to derive the supportive additional data were all approved by the respective human research ethics committee and were conducted in adherence with the Declaration of Helsinki, and written informed consent was obtained from all participants included.

Data Analyses

A statistical definition was applied to each statement to determine whether consensus was reached, as per previous recommendations[22] and as performed in 2 previous landmark Delphi studies that derived consensus-based clinical classification scales in our field.[23,24] Specifically, the median rating was calculated and interpreted as follows: ratings between 1 to 3 = disagreement, between 4 to 6 = neutral, and between 7 to 9 = agreement. Consensus was defined when ≥80% of the responses were within one of these above 3-point brackets containing the median.

Results

The results of the final round of the Delphi exercise are presented below, based on responses from 22 participants who discussed the results from the second round and completed the final survey. Table 2 summarizes the responses to 3 statements about the appropriateness of the onset of end-stage atrophic AMD as a clinical end point for early intervention trials. Consensus was reached in agreement that onset of end-stage atrophic AMD is an appropriate clinical end point for such trials (82% of ratings between 7 to 9). The highest level of disagreement (59% with ratings between 1 to 3) was noted for the statement that such an end point is inappropriate as it was too early and that direct measures of how a patient feels (e.g., from patient-reported outcome measures) or of everyday function (e.g., mobility performance) should be used instead.

Table 3 summarizes the responses to 5 statements about how an end point representing end-stage atrophic AMD should be defined in principle, broadly based on anatomical changes alone, anatomical changes associated with visual function loss shown in prior clinical studies, or functional loss that is directly measured during the course of a trial. Consensus was reached in agreement with the statement that end-stage atrophy should be defined based on anatomical changes shown in prior clinical studies to be associated with marked, but not complete, functional loss (95% with ratings between 7 to 9). The highest level of disagreement (68% with ratings between 1 to 3) was observed for the statement that end-stage atrophy should be defined based on anatomical changes without requiring evidence from previous clinical studies that they are associated with functional loss.

Table 4 summarizes the responses to 5 and 3 statements on the location and depth of functional loss required, respectively, if functional loss is to be measured when defining an end point representing end-stage atrophic AMD. Consensus was reached in disagreement with the statement that such functional loss should be located anywhere in the retina (86% with ratings between 1 to 3), and the highest level of agreement was noted for the statement that such functional loss should be located ≤500 μm from the foveal center (68% with ratings between 7 to 9). Consensus was reached in agreement that functional loss should be defined by a repeatable defect that was ≤10 dB (91% with ratings between 7 to 9), while consensus was reached in disagreement that such functional loss should be a repeatable defect <0 dB (82% with ratings between 1 to 3).

Table 5 summarizes the response to 15 statements regarding the frequency and severity of functional loss that should be observed in association with anatomical changes in clinical studies if such anatomical changes were used to define an end point representing end-stage atrophy. Consensus was not reached on any of these statements, but the highest levels of agreement were observed for the statements that ≥90% and ≥95% of lesions with such anatomical changes should have a repeatable ≤10 dB defect

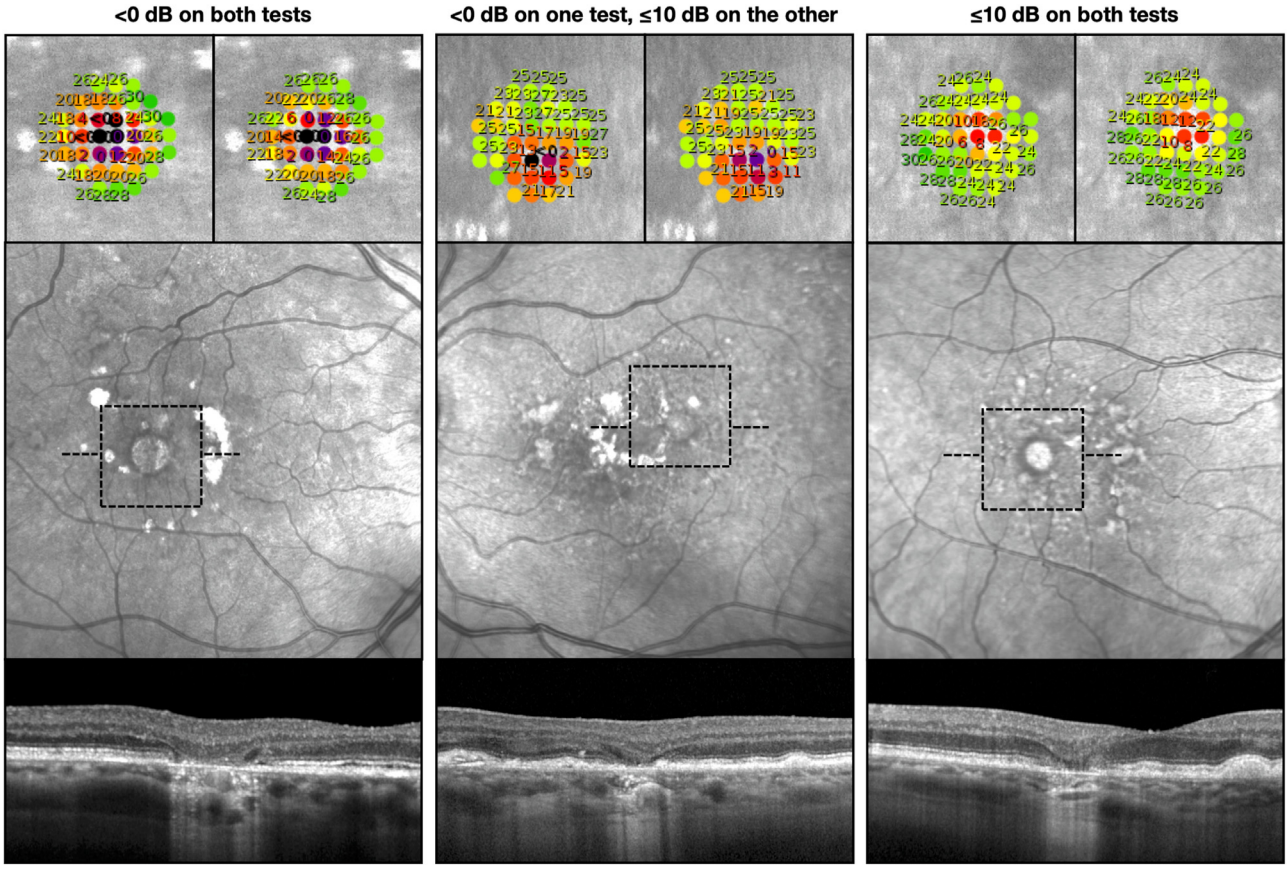


Figure 1. Three examples illustrating the different specific functional criteria evaluated in this study, based on the results from repeated targeted, high-density microperimetry tests (top row), with visual sensitivity thresholds (in decibels [dB]) at each location shown. The first criterion (left) required ≥1 location that was <0 dB (or an absolute scotoma) to be present on both tests, the second criterion (middle) required ≥1 location to be <0 dB on 1 test and ≤10 dB (marked, but not complete, loss of visual sensitivity) on the other test, and the third criterion (right) required ≥1 location to be ≤10 dB on both tests. The location where targeted, high-density microperimetry testing was performed is shown on the near-infrared reflectance image (middle row; indicated by the dashed square). The OCT B-scan through the region tested is shown for interest (bottom row; its location is indicated by the dashed horizontal lines on the near-infrared reflectance image).

when assessed in clinical studies (77% and 73% of ratings between 7 to 9). In contrast, the highest levels of disagreement were noted for the statements that ≥60% to ≥95% of lesions should have a repeatable <0 dB defect when assessed in clinical studies (between 64% and 68% of ratings between 1 to 3).

Discussion

This study showed that the CAM group reached consensus that the onset of end-stage atrophic AMD is an appropriate clinical trial end point for early intervention trials, and that such an end point should be defined based on anatomical changes shown in prior clinical studies to be associated with marked (but not necessarily complete) functional loss. Consensus was nearly reached for the criteria that ≥90% of such end-stage atrophic lesions should have ≥1 test location that is ≤10 dB on 2 tests based on a widely used clinical microperimeter, when assessed in clinical studies. We believe these findings will assist regulatory authorities when considering possible end points for early intervention trials in AMD.

The finding of consensus on the appropriateness of the onset of end-stage atrophy as a clinical trial end point by an international expert group in this study is supported by previous findings that the onset of late AMD, including atrophic AMD, was considered one of the 2 most important clinical outcomes to individuals with AMD.[31] This finding also reflects the longstanding historical evaluation of atrophic AMD onset as a clinical end point in intervention trials seeking to slow or prevent its onset.[12–15] However, the onset of an atrophic end point has not yet been used for the registration of an intervention with regulatory authorities. The United States Food and Drug Administration (FDA) and European Medicines Agency have both previously explicitly expressed only that the change or rate of change in the extent of existing GA would in principle be acceptable as a clinical end point to evaluate the efficacy of new treatments.[20,32] The FDA has indeed approved 2 treatments for slowing GA growth based on this end point.[33,34] The FDA has also previously stated, "*On the*

Table 1. Percentage of Repeated Targeted, High-Density Microperimetry Tests Meeting the Different Functional Criteria from the Simulation Model and Clinical Cohort

	Functional Criteria		
	<0/<0 dB	*<0/≤10 dB*	*≤10/≤10 dB*
Simulated Tests*			
True-positive rate	60%	87%	91%
False-positive rate	0.04%	0.4%	1.4%
Clinical cohort†			
GA on CFP			
≥175 μm	17%	33%	82%
≥500 μm	27%	52%	85%
HypoAF on FAF			
≥175 μm	12%	20%	60%
≥500 μm	24%	40%	81%
cRORA on OCT			
≥250 μm	10%	20%	58%
≥500 μm	34%	61%	90%

CFP = color fundus photography; cRORA = complete retinal pigment epithelium and outer retinal atrophy; dB = decibels; FAF = fundus autofluorescence; GA = geographic atrophy; HypoAF = hypoautofluorescence. <0/<0 dB = ≥1 location that was <0 dB on both tests; <0/≤10 dB = ≥1 location that was <0 dB on 1 test and ≤10 dB on the other test; ≤10/≤10 dB = ≥1 location that was ≤10 dB on both tests.
*As reported in Wu *et al*, *Invest Ophthalmol Vis Sci*; 2024.
†In the cohort reported in Wu *et al*, *Ophthalmol Sci*, 2024.

Table 2. Results for the Appropriateness of Onset of End-Stage Atrophic AMD as a Clinical End Point for Early Intervention Trials

Appropriateness of Onset of End-Stage Atrophic AMD as a Clinical Trial End Point	Percentage of Ratings between		
	1–3 (Disagree)	*4–6 (Neutral)*	*7–9 (Agree)*
Appropriate	9	9	82*
Inappropriate, as it is too late	36	32	32
Inappropriate, as it is too early	59	23	18

AMD = age-related macular degeneration.
*Consensus, based on ≥80% responses within a 3-point bracket containing the median.

question of whether GA onset might be acceptable as a potential trial end point, [the FDA] noted that any measurable change should be both statistically and clinically significant. Further, precisely defining the onset of GA would be crucial, given the variability seen in past studies. Surrogate end points, which predict a future clinically significant outcome, can be used in certain cases as the basis of approval of a new product."[19] The findings of this study therefore explicitly established consensus that the onset of end-stage atrophy would be acceptable as an end point when evaluating early interventions.

This study found that consensus was reached that end-stage atrophy should be defined not only using anatomical features, but also more specifically be based on anatomical features that have been shown in prior clinical studies to be associated with marked—but not necessarily complete—functional loss. These views differ from those of the FDA, which has stated, "*[GA] is a breakdown in the [RPE] and subsequent overlying retinal tissue… [where] photoreceptors are often spared at the periphery of the lesions… [and] surround an area where there is complete destruction of the photoreceptors [or an area of nonseeing retina]. Reduction in the rate of progression of these areas of complete destruction of the photoreceptors [that can] be measured indirectly by fundus photography or autofluorescence… is an acceptable end point.*" [32] In other words, the FDA has accepted anatomical changes of GA alone as a clinical trial end point on the basis that regions of GA represent areas of complete photoreceptor loss and therefore nonseeing retina. The above CAM consensus views also differ from those expressed by the European Medicines Agency, which state, "*The use of GA area as a primary efficacy variable could in principle be acceptable… [but] it needs to be justified that the marker represents a valid surrogate measure for visual function. Such justification could be based partly on literature demonstrating the prognostic value of GA area on visual function, but there would also be a need for support by evidence from the registration studies showing at least a trend of a positive effect of treatment on functional parameters.*"[20] In other words, the European Medicines Agency has expressed a desire to see evidence of treatment-related functional benefit as directly measured during registration studies. In contrast, the consensus view in this study is that an anatomical end point showing prior evidence of association with marked functional loss would be acceptable.

Furthermore, consensus was reached that a repeatable defect did not need to be an absolute scotoma (i.e., <0 dB), but that a deep visual sensitivity defect of ≤10 dB was a sufficient level of functional loss as measured on clinical microperimetry to define an end point representing end-stage atrophic AMD. Consensus was nearly reached on the specific criteria that such anatomical features should be shown in clinical studies to have ≥1 test location that is ≤10 dB on 2 microperimetry tests in ≥90% of these lesions (77% in agreement, just below the 80% threshold for consensus). This view is supported by findings from a recent study, which was considered as part of this Delphi study, which showed that even regions with a truly nonresponding test location—based on responses derived from stimuli presented at the optic nerve head—only had 1 test location that was ≤10 dB on 2 clinical microperimetry tests in 90% of the test pairs.[28] These previous findings likely reflect the subjective nature of visual function testing, where false-positive responses can occur, and mean that even locations that are truly nonresponding would not consistently be measured as having nonresponse. However, such an absence of consistent nonresponse could also be due to sampling errors with large fixation drifts that are not fully compensated by the fundus-tracking of clinical microperimeters (often ≥25 frames per second), or due to ocular stray light effects with bright stimuli. Nonetheless, support for the requirement of marked, but not necessarily complete, functional loss in this study makes provisions for these

Table 3. Results for How an End Point in a Clinical Trial Representing the Onset of End-Stage Atrophic AMD Should Be Defined

Definition of an End Point for End-Stage Atrophic AMD	Percentage of Ratings between		
	1–3 (Disagree)	*4–6 (Neutral)*	*7–9 (Agree)*
Anatomical changes regardless of association with functional loss*	68	9	23
Anatomical changes associated with complete functional loss*	23	18	59
Anatomical changes associated with marked functional loss*	5	0	95‡
Complete functional loss, as directly measured†	50	18	32
Marked functional loss, as directly measured†	23	0	77

AMD = age-related macular degeneration.
*As shown in prior clinical studies.
†During the course of a trial, such as based on microperimetry testing, where complete functional loss would be defined by nonresponse.
‡Consensus, based on ≥80% responses within a 3-point bracket containing the median.

factors that contribute to measurable responses in test locations that are truly nonresponding with clinical microperimetry testing. The views on the level of functional loss required may differ when using different parameters or methods for perimetry testing than the one considered in this Delphi study (e.g., a different stimulus pattern, smaller stimuli, or adaptive optics microperimetry[35]). This study considered clinical microperimetry testing based on Goldmann Size III stimuli since the FDA has previously accepted a visual field end point (and surrogates that are highly correlated with visual field loss) based on testing with this stimulus size when considering end points for optic neuropathies.[36–38]

Note that a Delphi study is intended to understand if convergence of opinion can be reached on a specific issue from domain experts.[21] It thus provides one source of information for regulatory authorities to consider along with other evidence when determining new clinical trial end points. However, a limitation of this Delphi study is that it was undertaken by a small subset of international AMD experts that were part of the CAM group, whose membership was overseen by 3 chairs.[5] Many of the participants in this Delphi study have also had longstanding preexisting relationships over many years. This may result in having a group that is composed of individuals whose opinions are more likely to be similar to each other than a group who is selected based on a quantitative criterion. Therefore, it is possible that the results from a Delphi study involving a different and/or larger group may result in different conclusions, and thus replication of this study with such a different and/or larger group will be invaluable in light of the findings of this study. Another limitation of this study is the potential impact of the conflict of interests on the outcomes of this study. However, we confirmed that the ratings by those in the final round of the Delphi study did not differ significantly between those with and without a conflict of interest (*data not shown*). Nonetheless, in addition to providing regulatory authorities with invaluable information to assist with their consideration of new clinical trial end points, the findings of this study also lay the foundation for subsequent clinical studies to identify the anatomical changes that are associated with the consensus-based functional criteria described in this report. Providing a robust set of anatomical features that are associated with the depth of functional loss expected by this expert group will help define the incident atrophic end point

Table 4. Results for the Required Location and Depth of Functional Loss If It Was Used to Define a Clinical Trial End Point Representing the Onset of End-Stage Atrophic AMD

	Percentage of Ratings between		
	1–3 (Disagree)	*4–6 (Neutral)*	*7–9 (Agree)*
Location of functional loss for an end point			
≤500 μm from the foveal center	27	5	68
≤1500 μm from the foveal center	18	23	59
≤3000 μm from the foveal center	32	50	18
Within the vascular arcades	59	23	18
Anywhere in the retina	86†	9	5
Depth of functional loss for an end point*			
<0 dB on both tests	82†	4	14
<0 dB on 1 test and ≤10 dB on the other	45	41	14
≤10 dB on both tests	0	9	91†

AMD = age-related macular degeneration; dB = decibel.
*In a scenario where microperimetry testing was performed twice with the Macular Integrity Assessment (MAIA) device, using conventional Goldmann Size III (0.43o diameter) against a mesopic background (1.27 cd/m^2).
†Consensus, based on ≥80% responses within a 3-point bracket containing the median.

Table 5. Results for What Frequency and Severity of Functional Loss That Anatomical Changes Should Be Associated with in Clinical Studies for an End Point in a Clinical Trial Representing Onset of End-Stage Atrophic AMD

	Percentage of Ratings between		
	1–3 (Disagree)	*4–6 (Neutral)*	*7–9 (Agree)*
≥1 location that is <0 dB on both tests*			
Seen in ≥60% of lesions	55	18	27
Seen in ≥70% of lesions	64	14	23
Seen in ≥80% of lesions	64	23	14
Seen in ≥90% of lesions	64	23	14
Seen in ≥95% of lesions	68	18	14
≥1 location that is <0 dB on 1 test and ≤10 dB on the other*			
Seen in ≥60% of lesions	41	32	27
Seen in ≥70% of lesions	41	23	36
Seen in ≥80% of lesions	36	27	36
Seen in ≥90% of lesions	50	27	23
Seen in ≥95% of lesions	55	27	18
≥1 location that is ≤10 dB on both tests*			
Seen in ≥60% of lesions	27	32	41
Seen in ≥70% of lesions	27	27	45
Seen in ≥80% of lesions	18	14	68
Seen in ≥90% of lesions	5	18	77
Seen in ≥95% of lesions	23	5	73

AMD = age-related macular degeneration; dB = decibels.
*In a scenario where microperimetry testing was performed twice with the Macular Integrity Assessment (MAIA) device, using conventional Goldmann Size III (0.43° diameter) against a mesopic background (1.27 cd/m^2). Note that the sum of the percentages presented across the 3 rating categories may not add up to 100% due to rounding to a whole number.

that could potentially be considered acceptable for early intervention trials. Pilot data, based on image grading from an observational study undertaken by a single center, were considered in this Delphi study (and presented in this report) on how often atrophic AMD lesions of different sizes, as seen on different imaging modalities, have ≥1 test location ≤10 dB on 2 clinical microperimetry tests. Further work is needed to undertake a more comprehensive and robust evaluation of such structure–function associations to further inform the definition of an incident end-stage atrophic AMD end point.

In conclusion, this study showed that consensus was reached that the onset of end-stage atrophy is an appropriate clinical end point for early intervention trials, and that such an end point should be defined based on anatomical changes that are shown in prior clinical studies to be associated with evidence of marked, but not necessarily complete, functional loss. We believe these consensus findings will help with the process of defining incident clinical end points that are acceptable to regulatory authorities. Such incident end points will be invaluable to facilitate the evaluation of interventions earlier in the AMD disease process before irreversible vision loss due to the onset of end-stage atrophy.

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Abbreviations and Acronyms:
AMD = age-related macular degeneration; **CAM** = Classification of Atrophy Meetings; **dB** = decibels; **FDA** = Food and Drug Administration; **GA** = geographic atrophy; **RPE** = retinal pigment epithelium.

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